

Make notes Very detailed (full sentences, no short phrases)

- ✓ Including all important theoretical points
- ✓ Including procedural steps wherever steps are given
- ✓ Formatted cleanly for exam

make sure to include each and every imp point.



Below are **clean, very detailed, full-sentence exam-oriented notes** prepared carefully from **AWS Academy Cloud Foundations – Module 2: Cloud Economics and Billing**.

(All content is taken from the uploaded module slides  AcademyCloudFoundations_Module_...)

MODULE 2 – CLOUD ECONOMICS AND BILLING (DETAILED EXAM NOTES)

✓ SECTION 1: FUNDAMENTALS OF AWS PRICING

◆ Three Fundamental Drivers of Cost in AWS

AWS pricing is based mainly on three core components:

1. Compute Cost

Compute refers to the processing power used by services such as Amazon EC2.

- AWS charges compute based on time used, usually per second or per hour (Linux instances support per-second billing).
- The cost varies depending on the instance type, size, and performance capacity selected.

2. Storage Cost

Storage refers to saving data in services like Amazon S3, EBS, and EFS.

- AWS generally charges storage based on the number of gigabytes (GB) stored.
 - The longer and larger the storage usage, the higher the cost.
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3. Data Transfer Cost

- Outbound data transfer (data sent out of AWS) is usually charged and aggregated.
 - Inbound data transfer (data coming into AWS) is mostly free, with few exceptions.
 - Charges are typically calculated per GB transferred.
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♦ AWS Payment Philosophy

AWS follows a flexible and customer-friendly pricing model based on three principles:

✅ Pay for What You Use

- Customers only pay for the services they consume.
- There are no large upfront investments.
- You can start and stop resources whenever required.
- This removes the need for long-term contracts.

👉 Unlike on-premises infrastructure where hardware must be purchased in advance, AWS allows dynamic usage.

✅ Pay Less When You Reserve (Reserved Instances – RI)

Reserved Instances allow customers to commit to long-term usage (usually 1 or 3 years) in return for heavy discounts.

AWS offers three RI payment options:

1. All Upfront Reserved Instance (AURI)

- Entire payment is made in advance.
- Provides the highest discount (up to 75%).

2. Partial Upfront Reserved Instance (PURI)

- Some amount is paid in advance, rest monthly.
- Medium discount.

3. No Upfront Reserved Instance (NURI)

- No initial payment, only monthly billing.
- Smallest discount.

✅ Pay Less When You Use More (Volume-Based Pricing)

AWS offers tiered pricing for many storage services.

- As usage increases, the cost per GB decreases.
- Services such as Amazon S3, Amazon EBS, and Amazon EFS use this model.
- This helps large organizations save more at scale.

✅ Pay Even Less as AWS Grows

- AWS continuously lowers prices as technology improves.
- Savings from economies of scale are passed to customers.
- AWS has reduced prices many times since 2006.
- New high-performance resources replace old ones at no extra cost.

◆ Custom Pricing

- AWS provides special pricing for large-scale or unique business requirements.
- It is available mainly for high-volume enterprise projects.

AWS FREE TIER

AWS Free Tier allows new users to explore AWS services for free for one year.

Steps involved:

1. Create an AWS account
2. Learn through short tutorials

3. Start building applications on AWS

This helps beginners gain hands-on cloud experience without cost.

SERVICES WITH NO DIRECT CHARGE

Some AWS services do not charge for usage themselves, but the resources they create may incur costs.

Examples:

- Amazon VPC
 - AWS Elastic Beanstalk
 - AWS Auto Scaling
 - AWS CloudFormation
 - AWS Identity and Access Management (IAM)
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Key Pricing Takeaways

- Inbound data transfer is mostly free.
 - Data transfer within the same AWS region is free.
 - You only pay for resources used.
 - You can stop services anytime without penalties.
 - Some services are free, but the resources they launch may not be free.
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SECTION 2: TOTAL COST OF OWNERSHIP (TCO)

On-Premises vs AWS Cloud

Traditional Infrastructure includes:

- Purchasing servers and equipment
- Hiring staff for administration

- Paying for contracts and maintenance
- Managing facilities (space, power, cooling)

AWS Cloud provides:

- No upfront hardware cost
 - Pay-as-you-go model
 - Ability to scale up and down instantly
 - Self-service infrastructure
 - Faster deployment and agility
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What is Total Cost of Ownership (TCO)?

Total Cost of Ownership (TCO) is a financial method used to calculate:



Both direct and indirect costs of running an IT system.

Why TCO is used:

- To compare on-premises infrastructure with AWS cloud
 - To calculate business savings
 - To build a strong business case for cloud migration
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TCO Cost Components

1. Server Costs

Includes:

- Server hardware
 - Power distribution units
 - Switches
 - Maintenance
 - Software licenses
 - Data center space, power, and cooling
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2. Storage Costs

Includes:

- Storage disks
 - SAN/Fibre Channel equipment
 - Storage administration
 - Facility expenses
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3. Network Costs

Includes:

- Network switches
 - Load balancers
 - Bandwidth usage
 - Network administration
 - Facility costs
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4. IT Labor Costs

Includes:

- System administrators
 - Maintenance teams
 - Operational management
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Result of Cloud Migration (Example from Case Study)

Organizations moving to AWS experienced:

- Up to 96% yearly cost savings
 - Huge reduction in servers
 - Lower IT labor costs
 - Improved performance
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AWS Pricing Calculator

The AWS Pricing Calculator helps users:

- Estimate monthly cloud costs
 - Compare pricing options
 - Identify cost-saving opportunities
 - Model architecture before deployment
 - Select instance types and contract terms
 - Organize estimates into service groups
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Reading an AWS Cost Estimate

An estimate includes:

- First 12 months total cost
 - Total upfront payment
 - Monthly recurring cost
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Additional Benefits of AWS (Beyond Cost)

Hard Benefits:

- Reduced spending on hardware and networking
- Lower capital expenditure
- Reduced backup and disaster recovery cost
- Smaller operations staff

Soft Benefits:

- Higher developer productivity
 - Better customer satisfaction
 - Agile business operations
 - Faster innovation
 - Global scalability
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SECTION 3: AWS ORGANIZATIONS



What is AWS Organizations?

AWS Organizations is a service used to manage multiple AWS accounts centrally.



Key Features

- Policy-based account management
 - Group-based management using Organizational Units (OUs)
 - APIs to automate account creation and control
 - Consolidated billing across all accounts
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Security in AWS Organizations

IAM Policies:

- Control access for users, groups, and roles within an account.

Service Control Policies (SCPs):

- Control what AWS services can be used across accounts or OUs.
 - Act as permission boundaries.
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Steps to Set Up AWS Organizations

1. Create the organization
 2. Create organizational units (OUs)
 3. Create service control policies
 4. Test restrictions and access
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Access Methods

AWS Organizations can be accessed using:

- AWS Management Console
 - AWS CLI
 - SDKs
 - HTTPS APIs
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SECTION 4: AWS BILLING & COST MANAGEMENT

AWS provides centralized tools to monitor and control spending.

Main Tools:

- AWS Billing Dashboard
 - AWS Cost Explorer
 - AWS Budgets
 - AWS Cost and Usage Reports
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Monthly Bills

- Show detailed service-wise charges
 - Help track usage trends
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Cost Explorer

- Visualize spending patterns
 - Forecast future costs
 - Identify high-cost services
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Cost and Usage Reports

- Provide detailed usage data
 - Used for advanced analysis
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SECTION 5: AWS TECHNICAL SUPPORT

AWS provides technical help for:

- Testing and learning
 - Production workloads
 - Mission-critical systems
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Support Tools

- Technical Account Manager (TAM)
 - AWS Trusted Advisor (best practice recommendations)
 - AWS Support Concierge for billing help
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AWS Support Plans

1. Basic Support (Free)

Includes:

- Documentation
 - Forums
 - Service Health Dashboard
 - Trusted Advisor checks
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2. Developer Support

- For development and testing
 - Limited technical assistance
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3. Business Support

- For production workloads
 - Faster response times
 - Trusted Advisor full access
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4. Enterprise Support

- For mission-critical workloads
 - Dedicated TAM
 - Highest priority responses
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MODULE SUMMARY

- ✓ AWS pricing is usage-based and flexible
 - ✓ TCO helps compare cloud vs traditional IT
 - ✓ AWS Pricing Calculator estimates cloud cost
 - ✓ AWS Organizations simplifies multi-account billing
 - ✓ AWS Billing tools monitor expenses
 - ✓ AWS Support plans help at every business stage
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If you'd like, I can now:

- ✓ Convert this into **quick revision notes**
- ✓ Make **memory tricks**
- ✓ Create **5-mark & 10-mark exam answers**
- ✓ Provide **important questions**

Just tell me 👍

Sources

